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Amrita Vishwa Vidyapeetham

Amrita School of Engineering, Coimbatore

B.Tech. Second Assessment Examinations – Aug. 2018

Seventh Semester

Computer Science and Engineering

15CSE480 Internet of Things

[Time: Two hours Maximum: 50 Marks]

Answer all questions

PART A (6 x 5 =30 Marks)

1. What are the main goals of CoAP ? Explain the design requirements of CoRE.
2. How WebSocket differs from HTTP and AJAX ? Discuss about the design aspects of WebSocket to work well with the legacy web infrastructure.
3. Develop a NodeMCU and DHT11 based IoT application using any one of your favourite programming language to display the current state of a temperature, humidity and dew point in a web page.
4. What is meant by IFTTT? Design a scenario to make use of IFTTT for an IoT based industrial application.
5. What are the needs for integration of an IoT application with a cloud ? Discuss about the various technical challenges associated with the integration of IoT with Cloud.
6. List few popular industrial grade cloud service providers. Discuss about the various aspects to be considered for selection of Cloud for IoT based Healthcare Application.

PART B (2 x 10 =20 Marks)

7. In the context of CoAP, discuss in detail about the following:
   a) CoAP request and response message with typical examples
   b) Non-Conformal Request
   c) Handling of packet loss
   d) Proxying and Caching

8. Design an IoT based Smart Parking System integrated with any one cloud of your choice. (Public / Private). Sample System Description / Use-Case:

   Smart parking is a typical IoT application that can benefit from advances in sensor, actuator and RFID technologies to provide many services to its users and parking owners of a smart city. This system considers a smart parking infrastructure where sensors are laid down on the parking spots to detect vehicle presence and RFID readers are embedded into parking gates to identify vehicles and help in the billing of the smart parking. Both types of devices

are endowed with wired and wireless communication capabilities for reporting to a gateway where the situation recognition is performed. The sensor devices are tasked to play one of the three roles: (1) slave sensor nodes located on the parking spot to detect car presence/absence; (2) master nodes located at one of the edges of a parking lot to detect presence and collect the sensor readings from the slave nodes; and (3) repeater sensor nodes, also called “anchor” nodes, located strategically at specific locations in the parking lot to increase the coverage and connectivity of the wireless sensor network. End User Interface and Review and Report analysis should be also considered.

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